# Directed Containers as Comonoids in the Sequential Monoidal Structure on Containers

Extracted from a machine-checked development

**Abstract**

We define comonoids in the monoidal category of containers under sequential composition, and show that every directed container determines such a comonoid. Counit and comultiplication are container morphisms $\varepsilon : C \Rightarrow I$ and $\delta : C \Rightarrow C \triangleleft C$, and the comonoid laws are equations of container morphisms built from the sequential action on morphisms and the monoidal unitors and associator. Unfolding these laws recovers the directed-container axioms D1–D5. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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## 1 Introduction

The category **Cont** of containers and container morphisms carries a monoidal structure (**Cont**, $\triangleleft$, $I$) whose tensor is sequential composition: the extension of $C \triangleleft D$ is the composite of the extensions of $C$ and $D$, and $I$ is the unit container (a single shape with a single position). A *comonoid* in this monoidal category is a container $C$ together with morphisms

$$\varepsilon : C \Rightarrow I, \qquad \delta : C \Rightarrow C \triangleleft C$$

satisfying the two counit laws and coassociativity, all stated internally to **Cont**.

This note shows that every directed container—a container with root, sub-shape, and shift satisfying D1–D5—gives rise to such a comonoid. The point of the internal formulation is that the laws never mention extensions or endofunctors on types: they are equations between container morphisms. Upon unfolding, one recovers the directed-container axioms on the nose, with the same partition that appears at the comonad level:

- right counit is D3;
- left counit is D1 (shapes) and D2 (positions);

- coassociativity is D4 (shapes) and D5 (positions).

Thus the correspondence between directed containers and comonads on extensions is an artifact of the monoidal structure on **Cont**, not merely of passage through the functor category.

We work in a single universe of types. Proofs are indicated by the phrase "by lean."

**What this note claims.** Directed containers are $\triangleleft$-comonoids. Explicitly: the root supplies a counit $C \Rightarrow I$; sub and shift supply a comultiplication $C \Rightarrow C \triangleleft C$; and D1–D5 are precisely the three comonoid laws for sequential composition. (The converse—that every comonoid arises this way—is not treated here.)

## 2 Background

**Definition 2.1** (Container morphism)**.** A **morphism** $\varphi : C \Rightarrow D$ of containers consists of a shape map $u : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$ and, for each shape $s$, a contravariant position map $g_s : \mathrm{Pos}(D)(u(s)) \to \mathrm{Pos}(C)(s)$.

**Definition 2.2** (Sequential composition)**.** The **sequential composite** $E \triangleleft C$ of containers has shapes $\sum_{t\,:\,\mathrm{Shape}(E)} \big(\mathrm{Pos}(E)(t) \to \mathrm{Shape}(C)\big)$ and positions over $\langle t,\, f\rangle$ given by $\sum_{u\,:\,\mathrm{Pos}(E)(t)} \mathrm{Pos}(C)(f(u))$. The unit container $I$ has a single shape and a single position.

We assume the monoidal structure on (**Cont**, $\triangleleft$, $I$): associator $\alpha$, left and right unitors $\lambda$ and $\rho$, and the action of $\triangleleft$ on morphisms (denoted $\triangleleft_2$ or $\mathrm{seq}_2$ when we need the binary map on arrows). Composition of container morphisms is written in diagrammatic order when convenient, so that $\delta; (\varepsilon \triangleleft C)$ means $(\varepsilon \triangleleft C) \circ \delta$ in applicative order.

**Definition 2.3** (Directed container)**.** A **directed container** is a container equipped with

$$\begin{aligned} \mathrm{o} &: (s : S) \to P(s), \\ \downarrow &: (s : S) \to P(s) \to S, \\ \oplus &: (s : S) \to (p : P(s)) \to P(s \downarrow p) \to P(s), \end{aligned}$$

satisfying axioms D1–D5 (sub under root; shift by root; shift by sub-root; nesting of subs; associativity of shift).

## 3 Technical lemmas on morphisms and transport

Because some directed-container axioms are only propositional equalities of shapes, comparing container morphisms whose shape maps agree only propositionally requires pointwise extensionality and a computation of position transport under sequential composition.

**Lemma 3.1** (Pointwise extensionality of morphisms)**.** *Let $f, g : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$ and let $f_p$, $g_p$ be position maps over $f$ and $g$ respectively. If $h_s : f(s) = g(s)$ for every $s$ and $f_p(s,p) = g_p\big(s,\, h_s \triangleright p\big)$ for every position $p$, then the morphisms $\langle f,\, f_p\rangle$ and $\langle g,\, g_p\rangle$ are equal.*

*Proof.* By lean. □

**Lemma 3.2** (Cancelation of cast round-trips)**.** *For a family $P$ and an equality $h : a = b$, transporting a value of $P(a)$ along $h$ and back along $h^{-1}$ returns the original value.*

*Proof.* By lean. □

**Lemma 3.3** (Sequential position transport)**.** *Let $E$ and $C$ be containers, let $t$ be a shape of $E$, and let $f, g : \mathrm{Pos}(E)(t) \to \mathrm{Shape}(C)$ be inner-shape assignments with $h_u : f(u) = g(u)$ for every outer position $u$. Transport of a position $\langle u,\, q\rangle$ of $E \triangleleft C$ over $\langle t,\, f\rangle$ along the induced shape equality yields*

$$\langle u,\ (h_u \triangleright q)\rangle \tag{1}$$

*over $\langle t,\, g\rangle$: the outer position is fixed and only the inner position is transported.*

*Proof.* By lean. □

# 4 Comonoids in $(\mathbf{Cont}, \triangleleft, I)$

**Definition 4.1** (Comonoid)**.** A **comonoid** in (**Cont**, $\triangleleft$, $I$) consists of a container $C$ together with morphisms

$$\varepsilon : C \Rightarrow I, \qquad \delta : C \Rightarrow C \triangleleft C, \tag{2}$$

such that the following equations of morphisms hold:

$$\delta;(\varepsilon \triangleleft C) \;=\; \lambda_C^{-1} \qquad \text{(left counit)}, \tag{3}$$

$$\delta;(C \triangleleft \varepsilon) \;=\; \rho_C^{-1} \qquad \text{(right counit)}, \tag{4}$$

$$\delta;(\delta \triangleleft C);\alpha_{C,C,C} \;=\; \delta;(C \triangleleft \delta) \qquad \text{(coassociativity)}. \tag{5}$$

**Remark 4.2.** These are the standard comonoid axioms in a monoidal category, written with the sequential tensor and its coherence isomorphisms.

# 5 From directed containers to comonoids

**Definition 5.1** (Counit from a directed container)**.** Let $D$ be a directed container. The **counit** $\varepsilon : D \Rightarrow I$ is the container morphism with unique shape map and position map

$$\varepsilon_p(s,\, *) \;=\; \mathrm{o}(s). \tag{6}$$

**Definition 5.2** (Comultiplication from a directed container)**.** The **comultiplication** $\delta : D \Rightarrow D \triangleleft D$ is the container morphism with

$$\delta_S(s) \;=\; \langle s,\ \lambda p.\, s \downarrow p\rangle, \tag{7}$$

$$\delta_p(s,\, \langle p,\, q\rangle) \;=\; p \oplus q. \tag{8}$$

**Theorem 5.3** (Directed containers are comonoids)**.** *For every directed container $D$, the morphisms $\varepsilon$ and $\delta$ of Definitions 5.1 and 5.2 satisfy the three comonoid laws of Definition 4.1. Thus $D$ is a $\triangleleft$-comonoid.*

*Proof.* By lean. □

**Remark 5.4** (How the axioms match)**.**

1. **Right counit.** Both sides have definitionally equal shape maps; the position equality is exactly D3.
2. **Left counit.** Shape equality is D1; after sequential position transport (Lemma 3.3), the position equality is D2.
3. **Coassociativity.** Shape equality is D4 (pointwise in outer and inner positions); position equality is D5 after cancelling a cast round-trip (Lemma 3.2) and applying sequential transport twice.

## 6 Discussion

**What has been established.** The directed-container structure is precisely the data of a comonoid in (**Cont**, $\triangleleft$, $I$), at least in the forward direction. The comonoid laws are equations of container morphisms; their unfolding is the classical list D1–D5. In particular, one need not pass through [**Set**, **Set**] to state or verify the comonoid equations.

**What is not established here.** The converse—that every $\triangleleft$-comonoid arises from a directed container—is not formalised in the source development. One expects it by the same representation and shape-collapse arguments that recover directed data from a container comonad on the extension, but that direction is left open.

**Machine checking.** Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

## Acknowledgements

The mathematical content follows the monoidal structure of sequential composition of containers and the directed-container theory of Ahman, Chapman, and Uustalu. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

## References

[1] D. Ahman, J. Chapman, and T. Uustalu. When is a container a comonad? *Logical Methods in Computer Science*, 10(3), 2014.

[2] M. Abbott, T. Altenkirch, and N. Ghani. Containers: constructing strictly positive types. *Theoretical Computer Science*, 342(1):3–27, 2005.